

B. VARSH Management

time limit per test: 1 second

memory limit per test: 256 megabytes

input: standard input

output: standard output

VARSH is optimizing the sorting facility for a new manufacturing plant. The facility features a large circular conveyor belt consisting of n discrete boxes (where n is an even integer), indexed from 0 to $n - 1$ in a clockwise direction.

There are two specialized workers stationed at the belt. Worker A is permanently stationed at box index 0 and strictly collects Waste Paper, denoted by the character **W**. Worker B is permanently stationed at the opposite end at box index $\frac{n}{2}$ and collects Plastic and Glass, denoted by the characters **P** and **G**.

The conveyor belt motor is currently restricted and can only rotate in the clockwise direction. Rotating the entire belt clockwise by exactly one box position consumes x units of energy. When a box containing waste arrives at a worker's index, the worker instantly removes all the waste they are responsible for at zero energy cost.

You are given the initial contents of all n boxes. Each box contains a string of characters representing the waste inside it. Empty boxes are represented by the character **.** (a period).

Find the minimum total energy the plant needs to spend to completely clear all **W**, **P**, and **G** waste from the conveyor belt.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10^4$). The description of the test cases follows.

The first line of each test case contains two integers n and x ($2 \leq n \leq 2 \cdot 10^5$; n is even; $1 \leq x \leq 10^9$) — the number of boxes and the energy cost per clockwise shift.

The next n lines describe the boxes. The i -th line contains a string s_i ($1 \leq |s_i| \leq \frac{n}{2}$) representing the waste in the i -th box. The string consists only of characters **W**, **P**, **G**, or **.**

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$, and the total length of all strings s_i over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single integer — the minimum energy required to clear all waste.